

CYBERNOVA TECHNOLOGIES

CAPSTONE PROJECT REPORT

Enterprise Identity and Access Management (IAM) Deployment Using Active Directory

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Domain	cybernover.local

1. Executive Summary

This report documents the successful deployment of a complete Active Directory (AD) environment for CyberNova Technologies. The solution includes a domain controller (DC01), a file server (FS01), and a client workstation (CLIENT01). Key implementations include departmental Organizational Units (OUs), security groups, user accounts, Group Policy Objects (GPOs) for password policy and access restrictions, and role-based file sharing. The solution meets all requirements for centralized identity management and secure access control.

2. System Architecture Overview

2.1 Virtual Lab Environment

Component	Specification
Virtualization Platform	Oracle VirtualBox
Host Operating System	Windows / Linux Host
Network Adapter Mode	Bridged Adapter

2.2 Virtual Machines

VM Name	Operating System	IP Address	Role
DC01	Windows Server 2022	172.16.1.10	Domain Controller, DNS Server
FS01	Windows Server 2022	172.16.1.20	File Server
CLIENT01	Windows 10/11	172.16.1.30	Domain Workstation

Domain Name: cybernova.local | **NetBIOS Domain Name:** CYBERNOVA

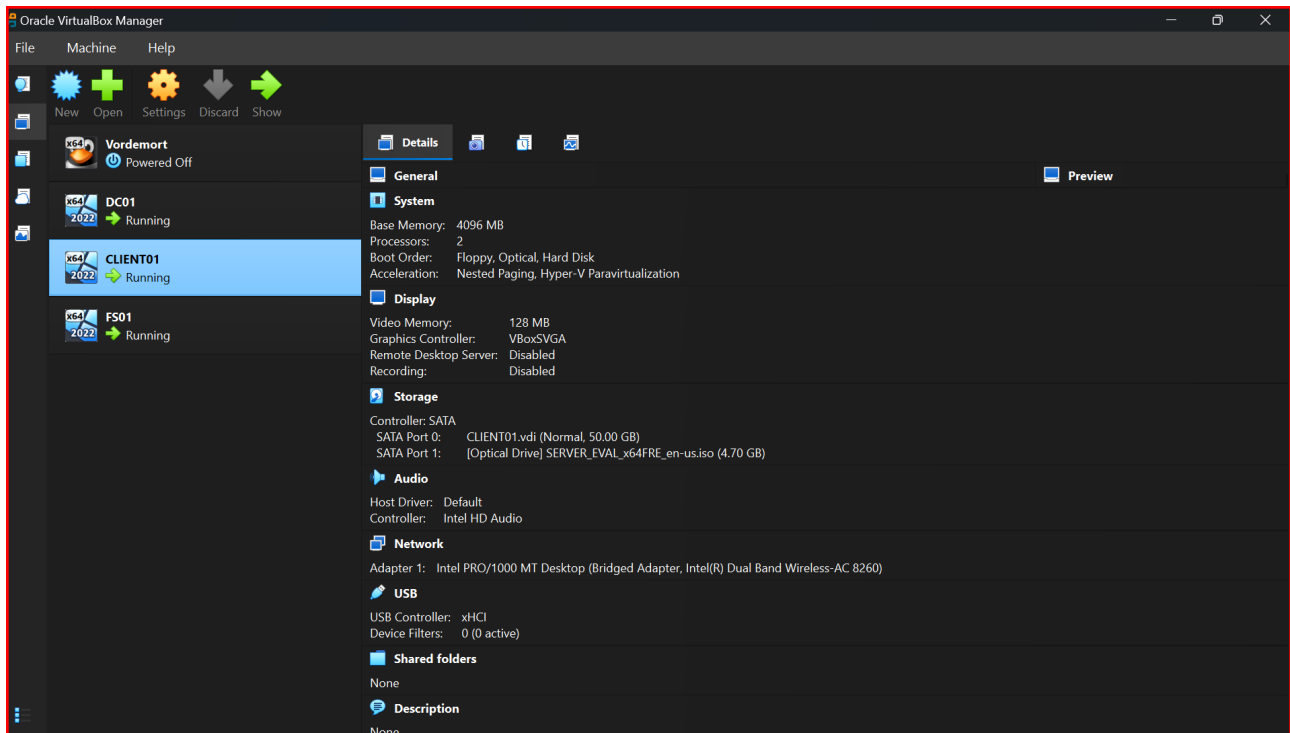


Figure 1: VirtualBox VM Environment Overview

3. Active Directory Deployment

3.1 Domain Controller Promotion

- Windows Server 2022 installed on DC01
- Active Directory Domain Services (AD DS) role installed
- Server promoted to Domain Controller
- New forest created: cybernova.local
- DSRM password configured

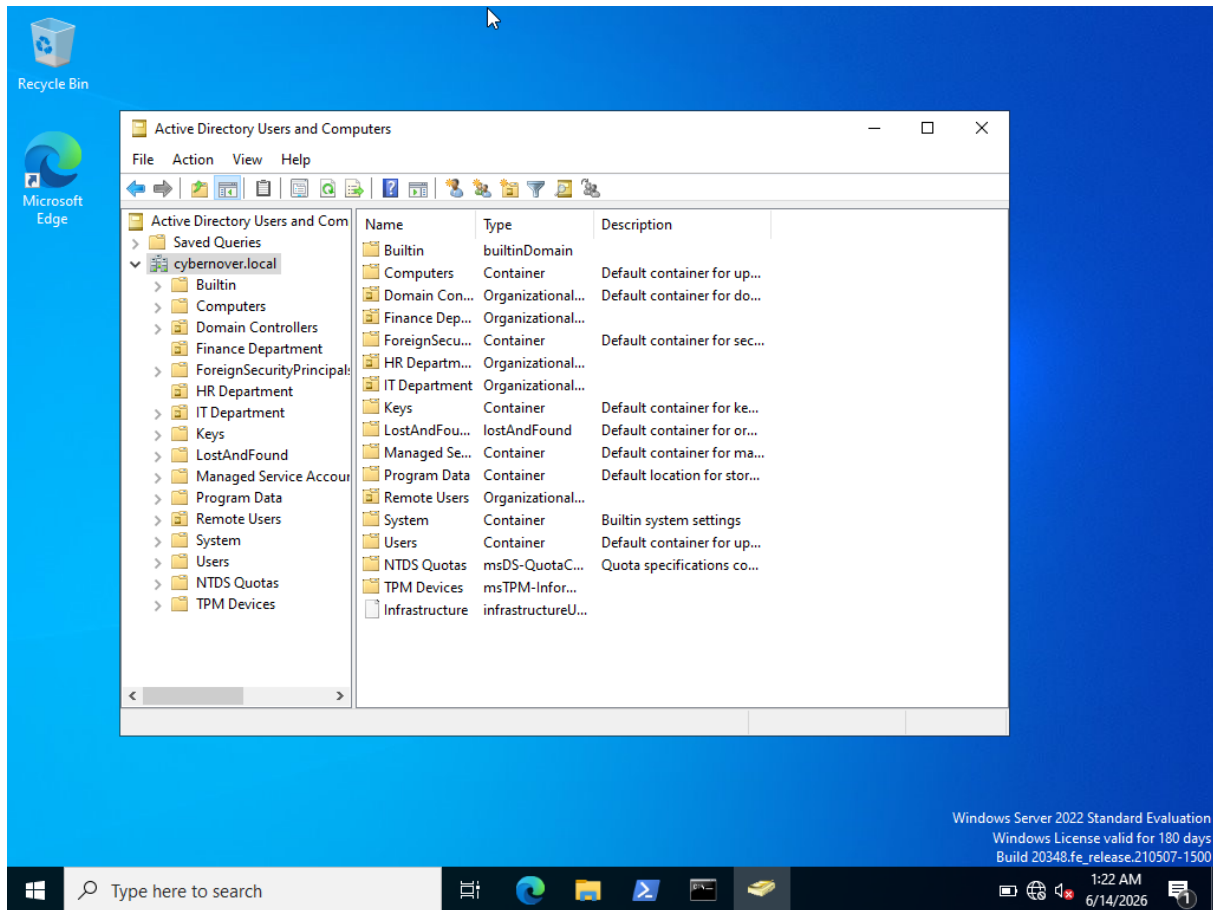


Figure 2: Domain Controller Promotion Wizard

4. Organizational Unit (OU) Structure

The following OUs were created at the domain root:

OU Name	Purpose
IT Department	Information Technology staff
HR Department	Human Resources staff
Finance Department	Finance and Accounting staff
Remote Users	Remote/Work-from-home employees

Justification: This department-based structure enables targeted GPO application, delegated administration, and easier management of user resources.

5. User and Group Management

5.1 Security Groups Created

Group Name	Department	Group Scope	Group Type
IT_Admns	IT Department	Global	Security
HR_Staff	HR Department	Global	Security
Finance_Staff	Finance Department	Global	Security
Remote_Users	Remote Users	Global	Security

5.2 User Accounts Created (9 Users)

Username	Full Name	Department	Group Membership
fadtech	IDDRISU FADEL	IT	IT_Admns
sandy.tech	Sandra Agyimang	IT	IT_Admns
lisa.hr	Lisa HRManager	HR	HR_Staff
david.hr	David HRCoord	HR	HR_Staff
anna.finance	Anna FinanceLead	Finance	Finance_Staff
paul.finance	Paul Accountant	Finance	Finance_Staff
tom.remote	Tom RemoteSales	Remote	Remote_Users
emma.remote	Emma RemoteSupport	Remote	Remote_Users

Password Policy: All users configured with password CyberNover2024! (Password never expires for lab environment)

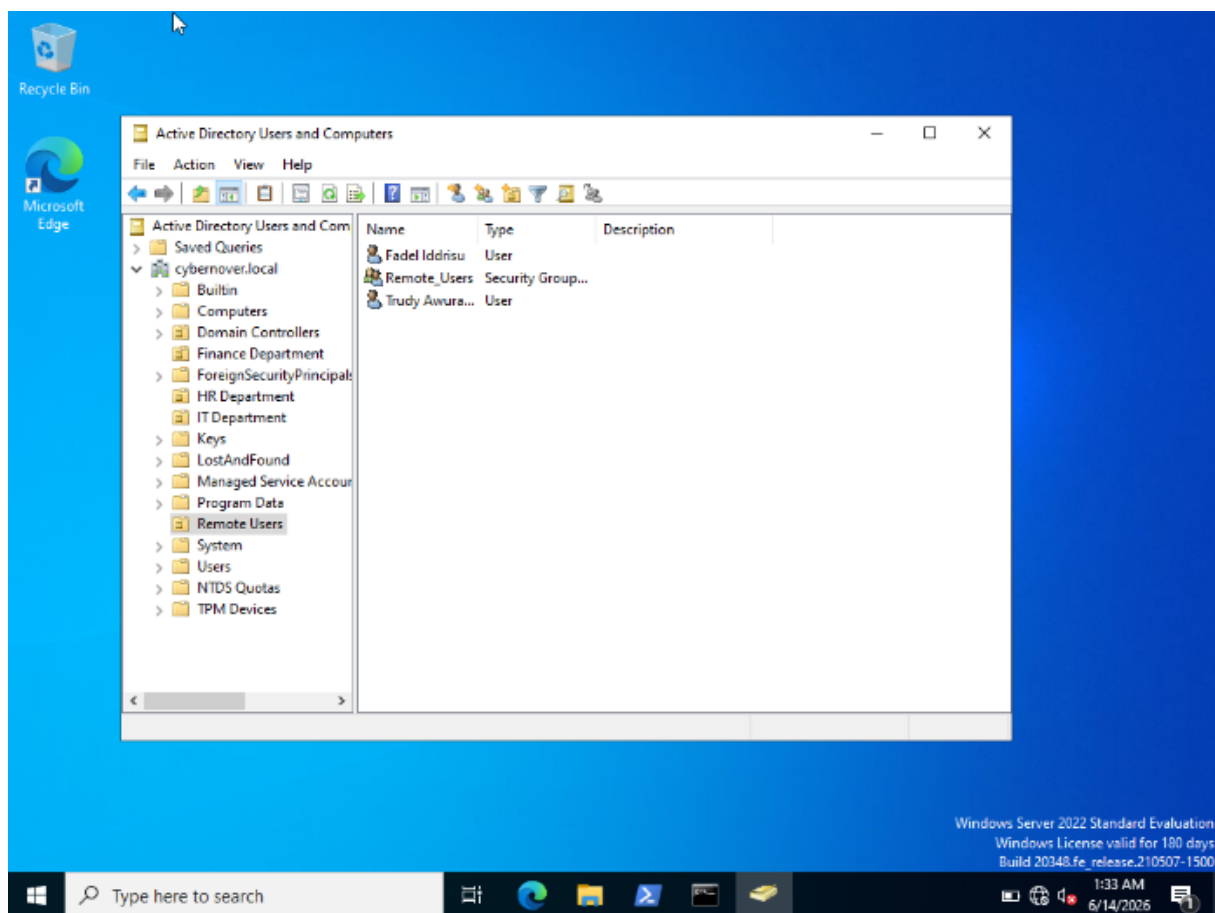


Figure 3: Security Groups in Active Directory

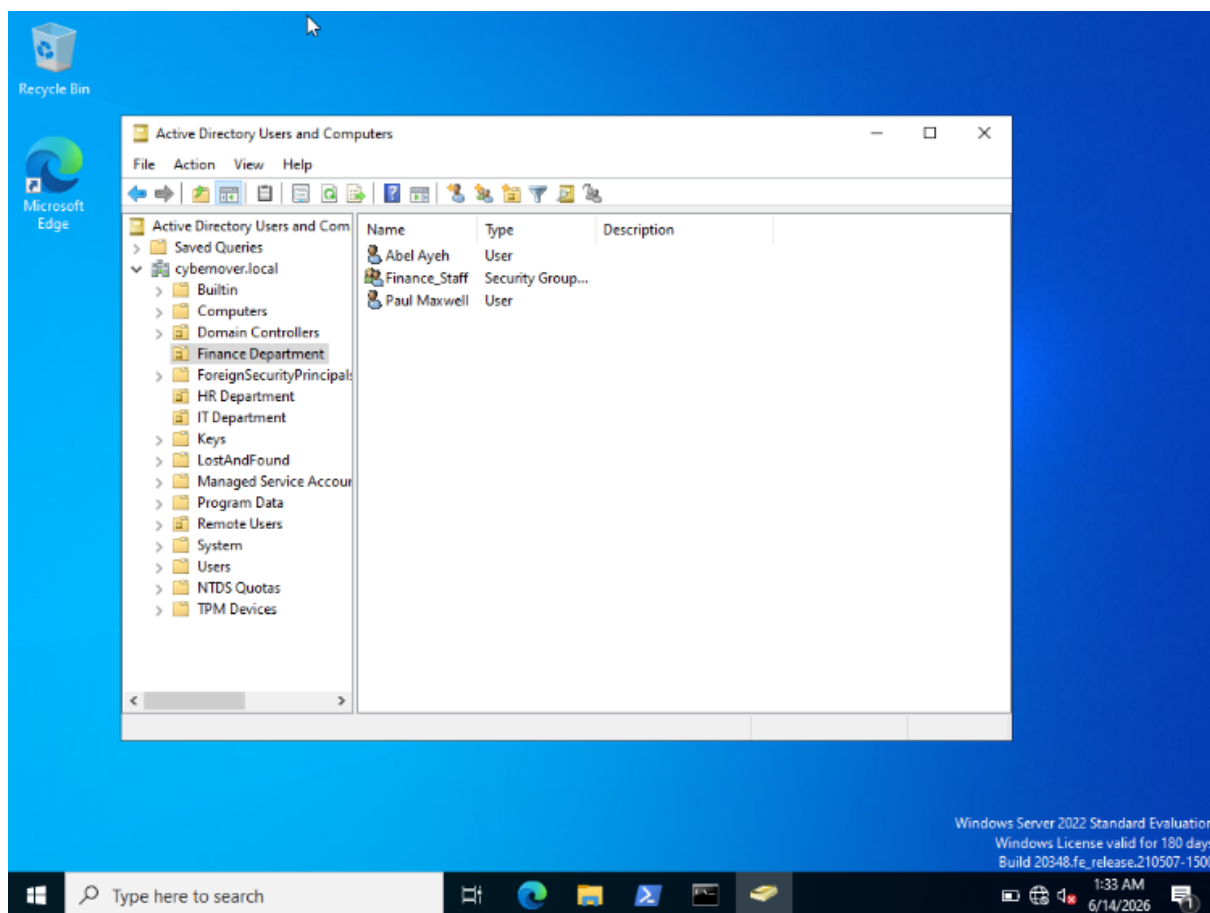


Figure 4: User Accounts Created

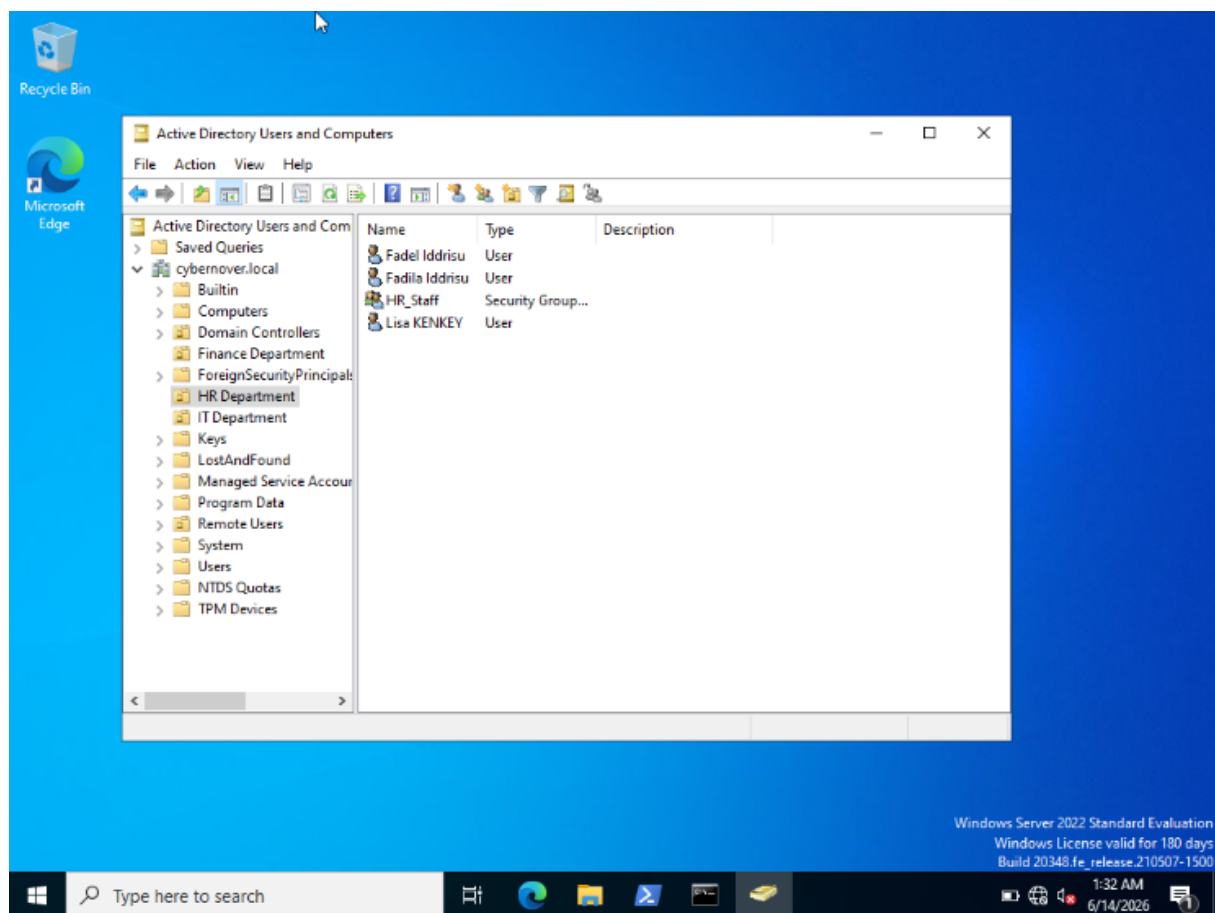


Figure 5: OU Structure in AD Users and Computers

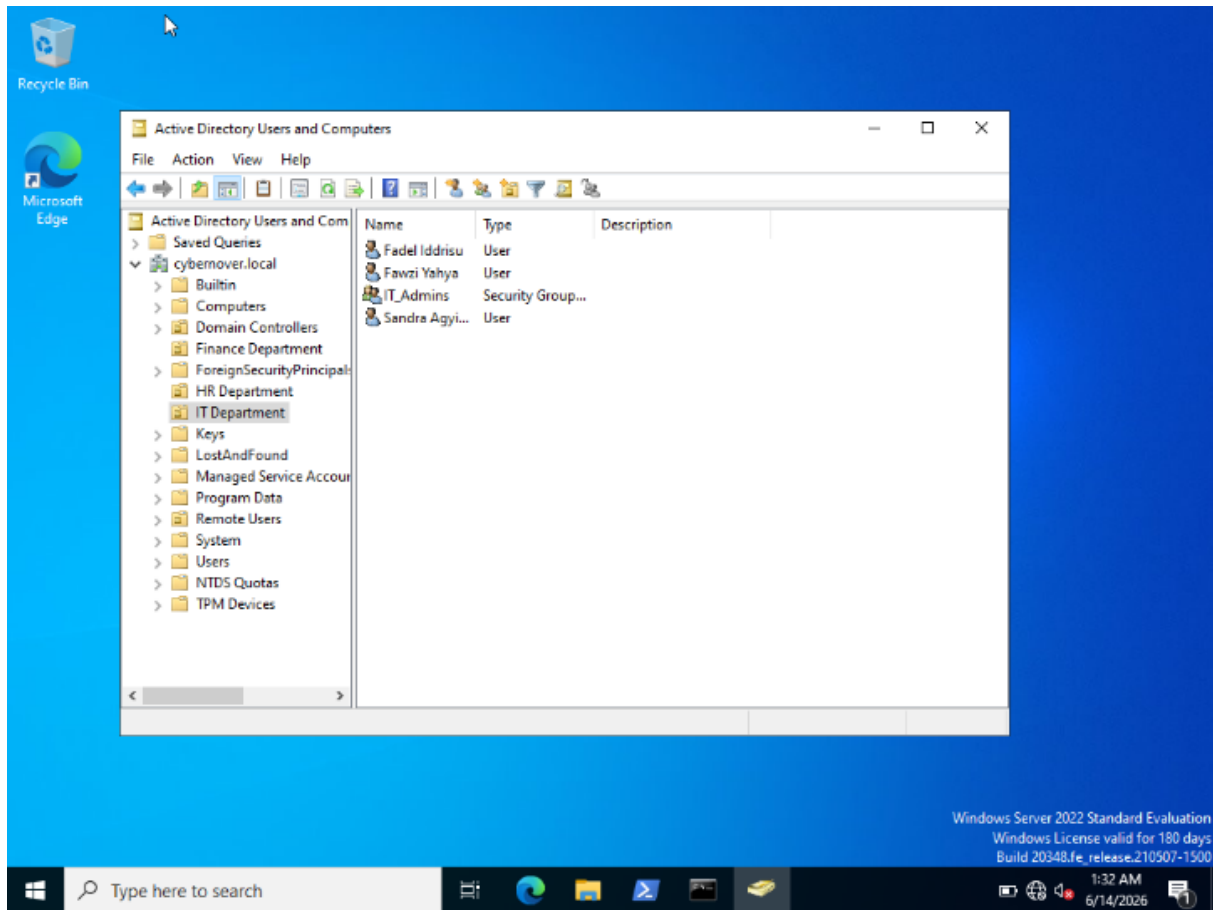


Figure 6: User Group Membership Configuration

6. Group Policy Object (GPO) Configuration

6.1 GPO 1: Default Domain Policy (Password & Lockout)

Location: cyberover.local (Linked by default)

Password Policy Settings:

Policy Setting	Configured Value
Minimum password length	8 characters
Password must meet complexity requirements	Enabled
Enforce password history	5 passwords remembered
Maximum password age	42 days
Minimum password age	1 day

Account Lockout Policy Settings:

Policy Setting	Configured Value
Account lockout threshold	5 invalid attempts
Account lockout duration	30 minutes
Reset account lockout counter after	30 minutes

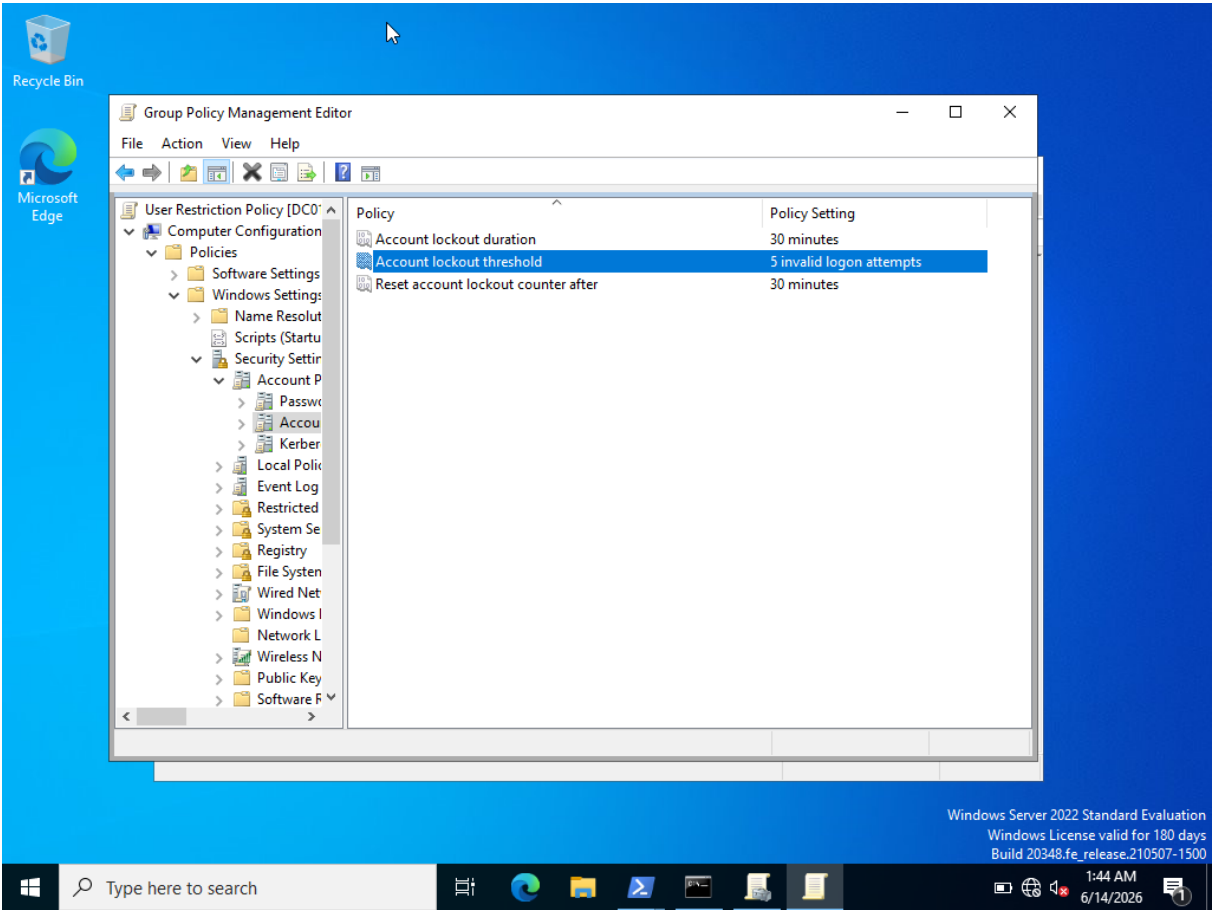


Figure 7: Password Policy GPO Settings

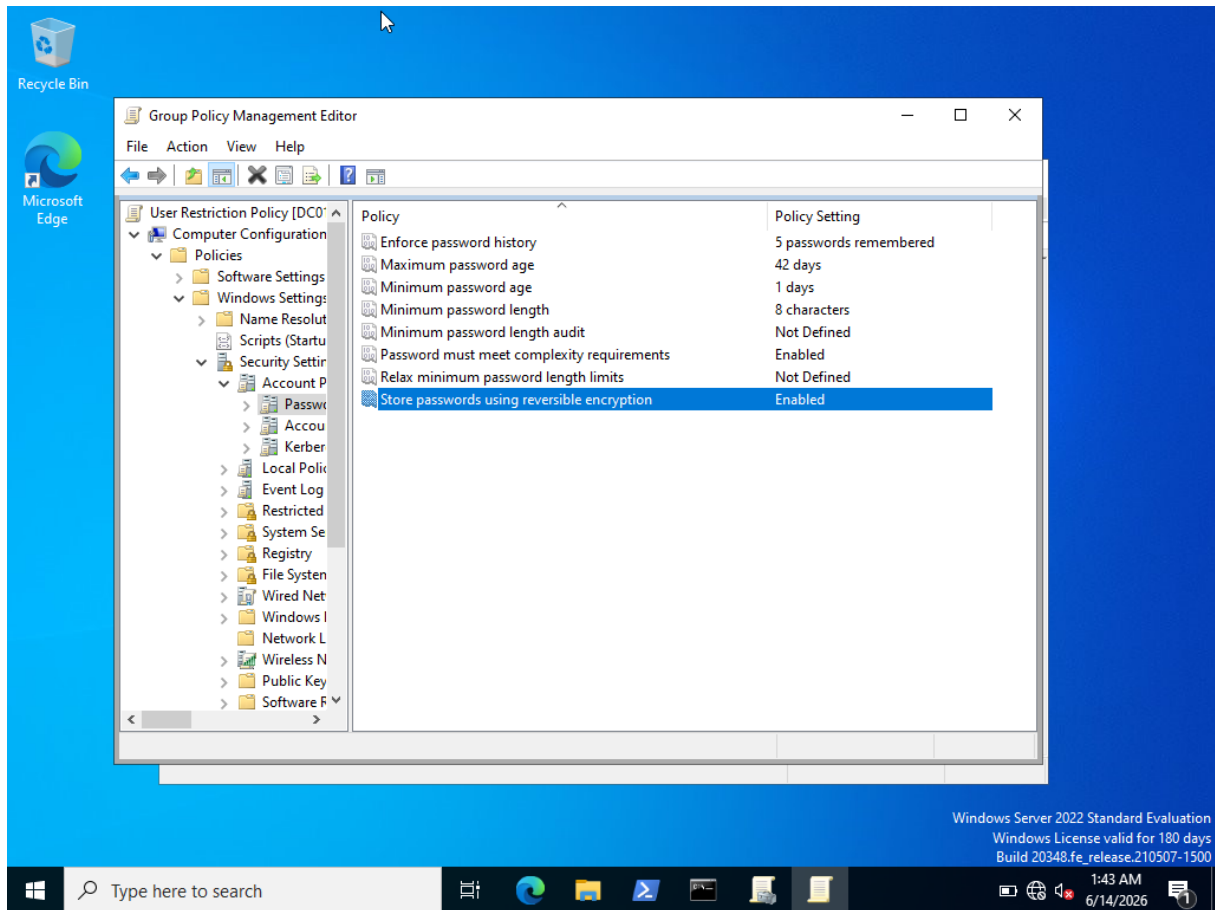


Figure 8: Account Lockout Policy Settings

6.2 GPO 2: User Restrictions Policy

Applied to OUs: IT Department, HR Department, Finance Department, Remote Users

Restriction	Configuration	Status
Control Panel access	Prohibited for standard users	■ Applied
Command Prompt access	Prevented for non-administrators	■ Applied
Screen lock after inactivity	900 seconds (15 minutes)	■ Applied
Screen saver password	Enabled	■ Applied

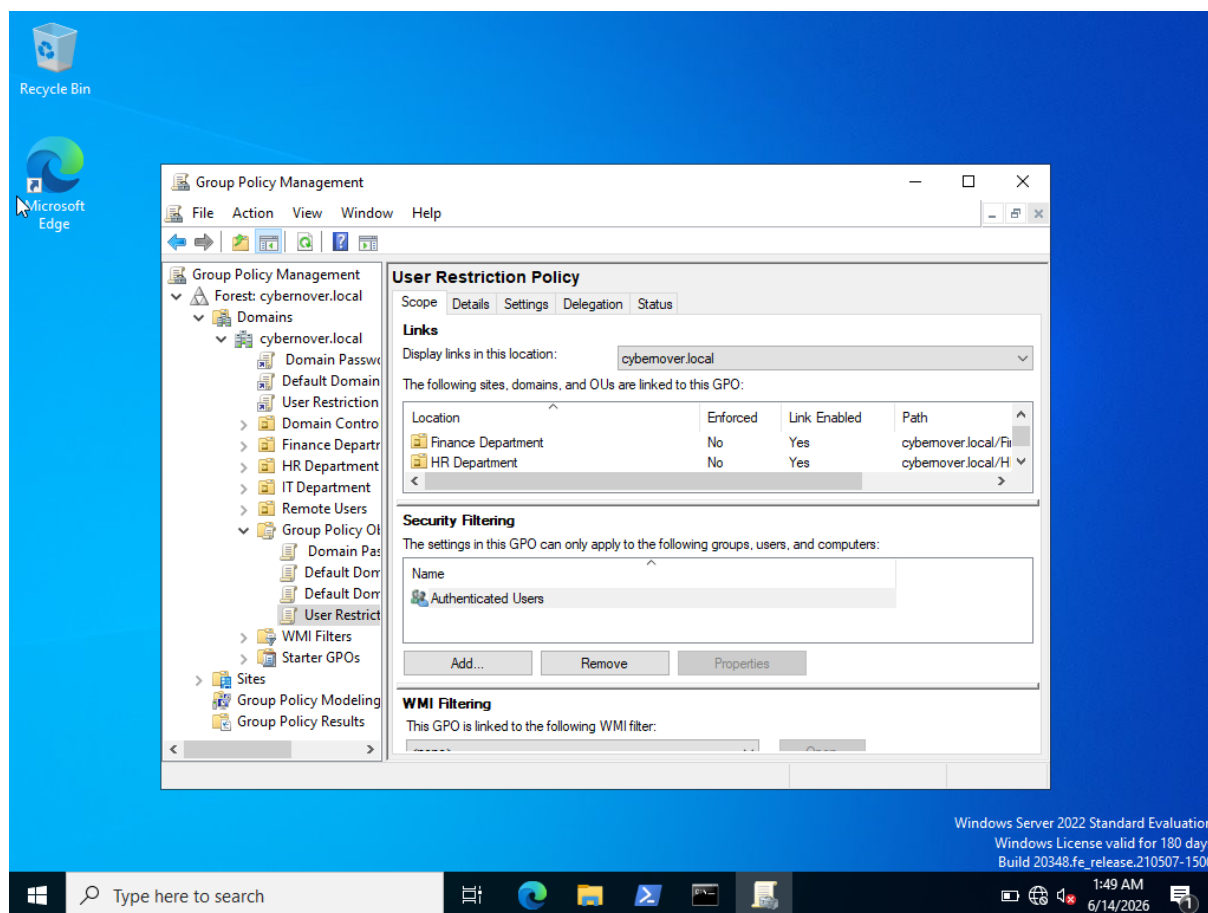


Figure 9: User Restrictions GPO – Control Panel

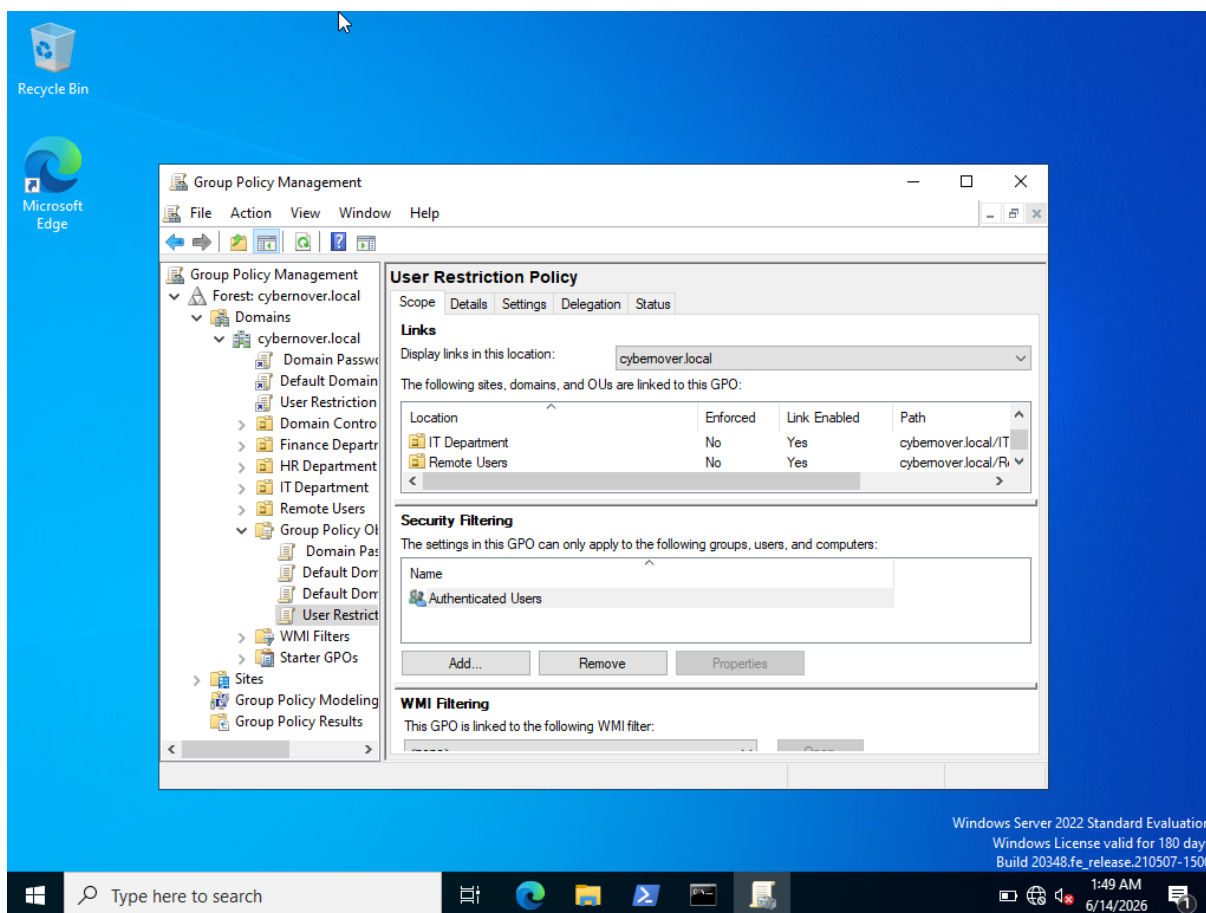


Figure 10: User Restrictions GPO – Command Prompt

7. File Sharing and Permissions

7.1 Shared Folders on FS01

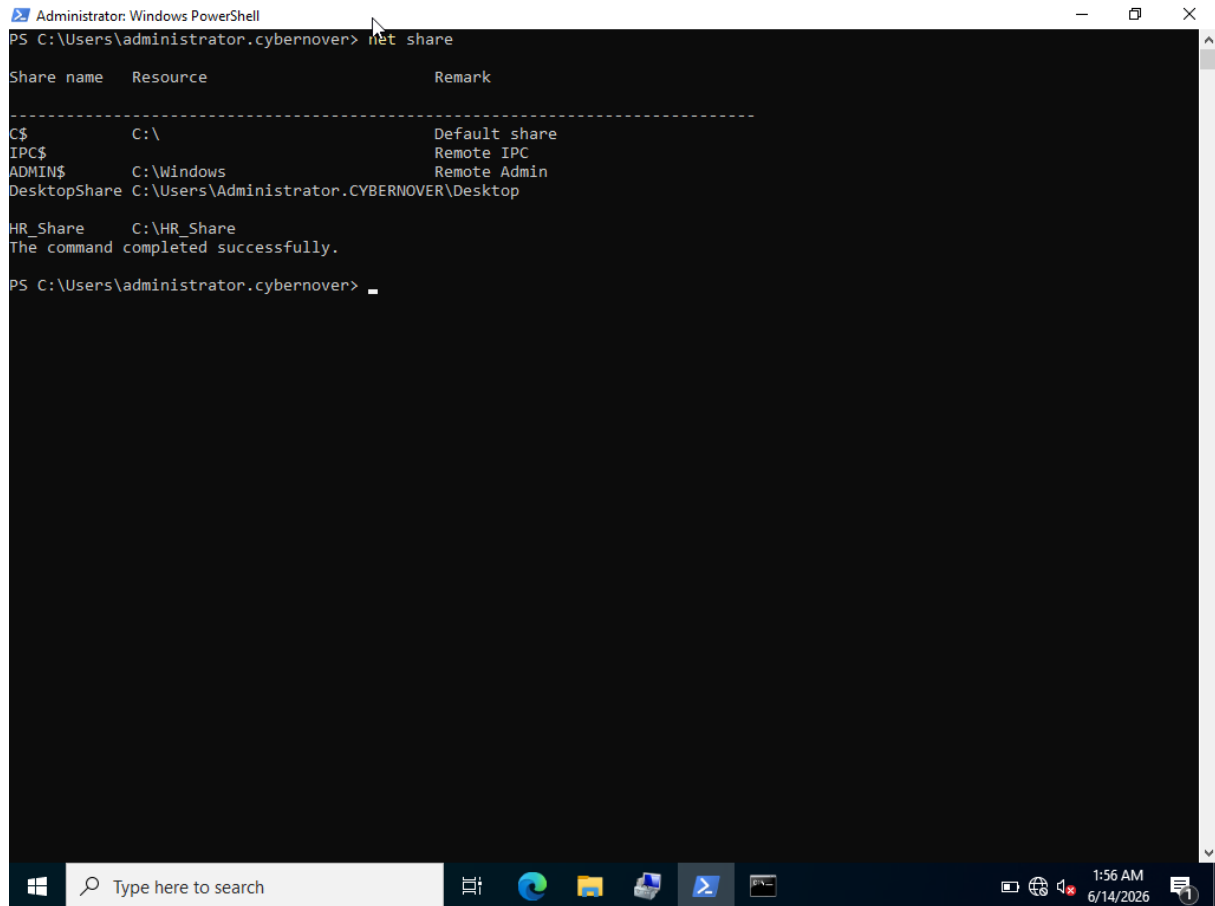
Share Name	Path	Access Group	Permission
IT_Share	C:\IT_Share	IT_Admins	Full Control
IT_Share	C:\IT_Share	Domain Users	Read
HR_Share	C:\HR_Share	HR_Staff	Modify
DesktopShare	C:\Users\Administrator.CYBERNOVER\Desktop	Finance_Staff	Read/Change

Note: Finance department access is provided via \\FS01\DesktopShare which contains the Finance_Share folder.

7.2 NTFS Permissions Configuration

Folder	Group	Permission	Inheritance
C:\IT_Share	IT_Admins	Full Control (F)	Disabled
C:\IT_Share	Domain Users	Read & Execute (RX)	Disabled

Folder	Group	Permission	Inheritance
C:\HR_Share	HR_Staff	Modify (M)	Disabled
Desktop	Finance_Staff	Read & Execute (RX)	Disabled



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Administrator: Windows PowerShell
PS C:\Users\administrator.cybernova> net share

Share name      Resource                Remark
-----
C$              C:\                    Default share
IPC$            C:\                   Remote IPC
ADMIN$          C:\Windows             Remote Admin
DesktopShare    C:\Users\Administrator.CYBERNOVER\Desktop
HR_Share        C:\HR_Share
The command completed successfully.
PS C:\Users\administrator.cybernova>

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Figure 11: Shared Folders on FS01

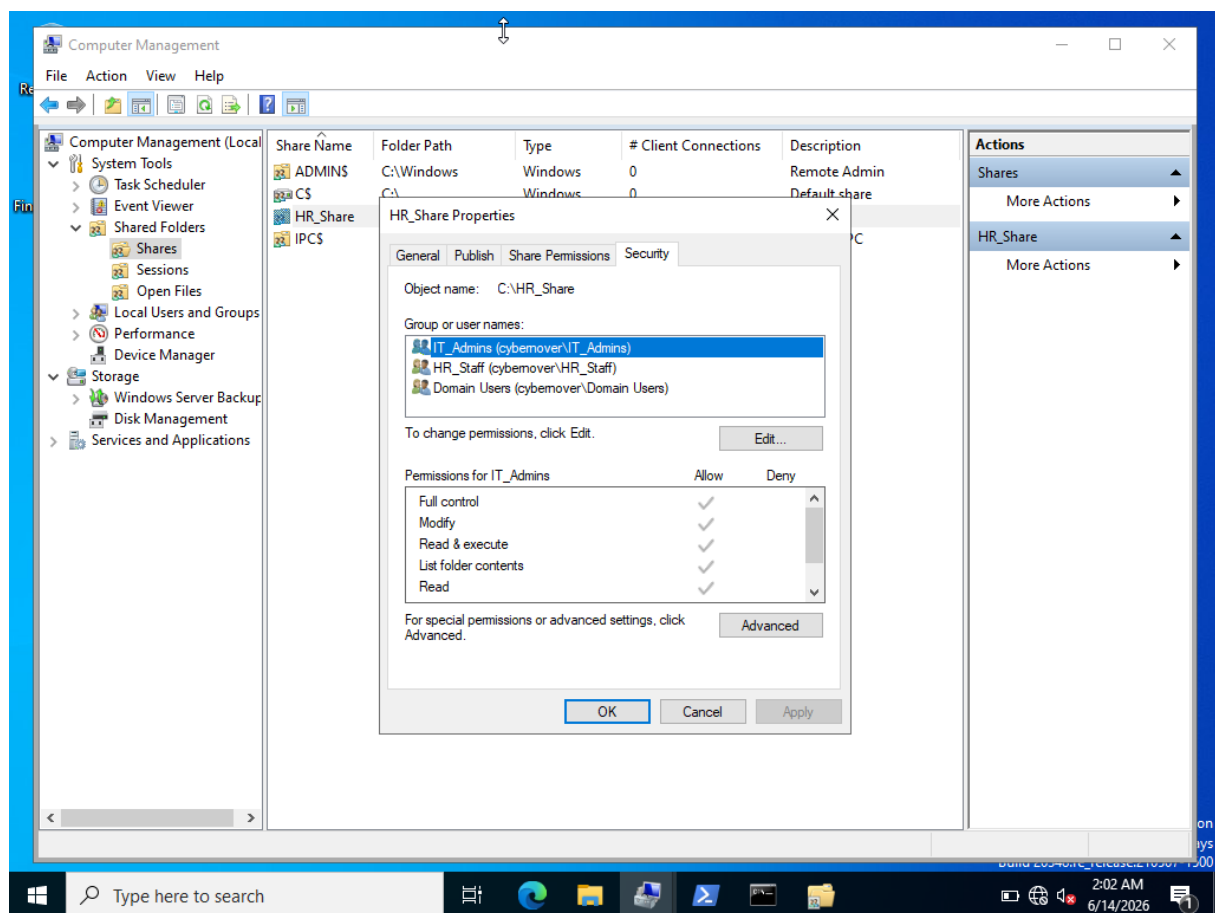


Figure 12: IT_Share NTFS Permissions

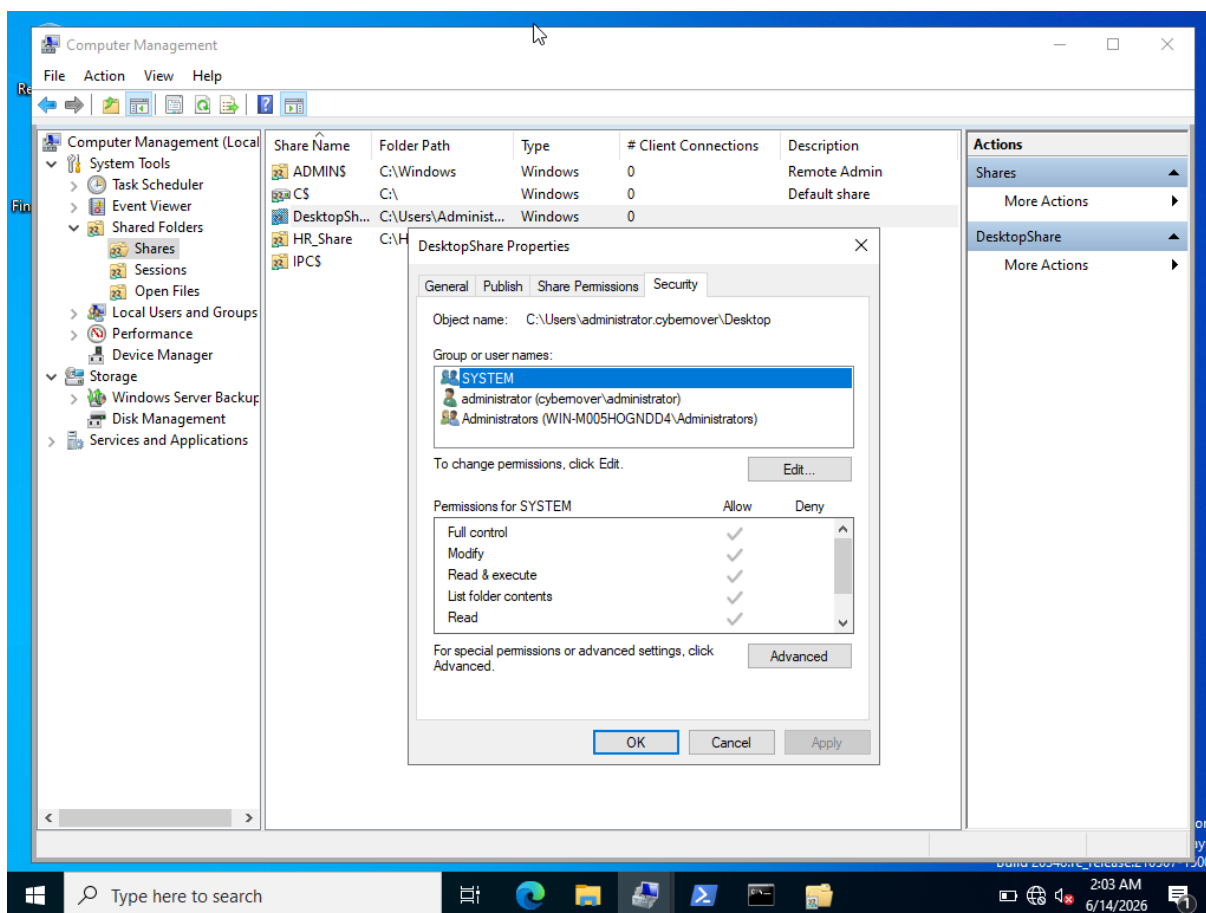


Figure 13: HR_Share Permissions Configuration

8. Access Testing Results

8.1 Test Matrix

User	IT_Share	HR_Share	DesktopShare (Finance)	Cmd Prompt	Control Panel
john.it (IT)	■ Access	■ Denied	■ Denied	■ Blocked	■ Blocked
lisa.hr (HR)	■ Denied	■ Access	■ Denied	■ Blocked	■ Blocked
anna.finance (Finance)	■ Denied	■ Denied	■ Access	■ Blocked	■ Blocked

Principle of Least Privilege: Users only have access to resources required for their department role. All GPO restrictions are successfully enforced.

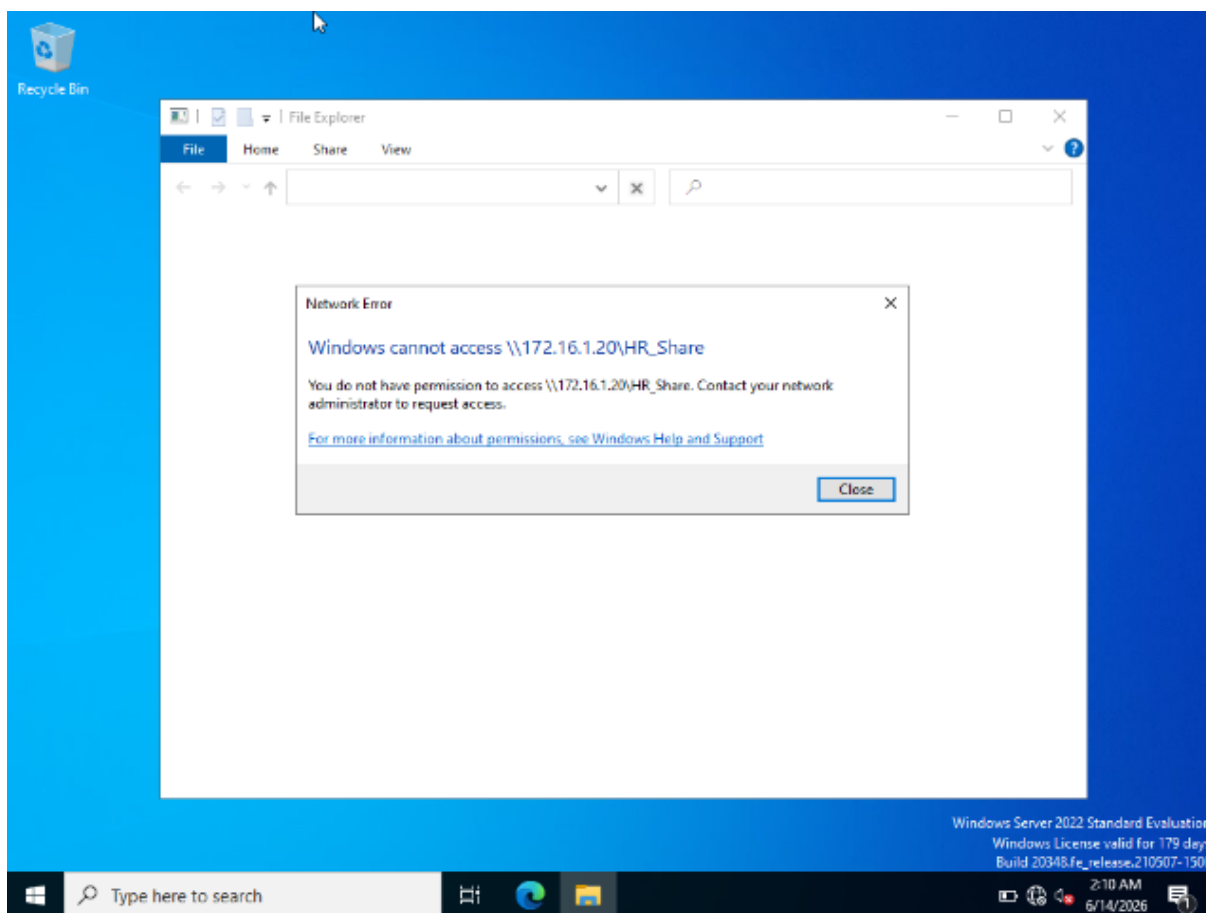


Figure 14: Access Testing Verification

9. Security Hardening

Requirement	Implementation	Status
Strong password policy	Enforced via Default Domain Policy GPO	■
Account lockout policy	Enforced via Default Domain Policy GPO	■
Control Panel restriction	Enforced via User Restrictions Policy GPO	■
Command Prompt restriction	Enforced via User Restrictions Policy GPO	■
Automatic screen lock	Enforced via User Restrictions Policy GPO (900s)	■
Windows Firewall	Enabled with file sharing rules	■
Auditing policies	Configured for logon events	■

11.1 User Creation Process

- ## 11.2 Password Reset Procedure

- ## 11.3 Resource Access Management

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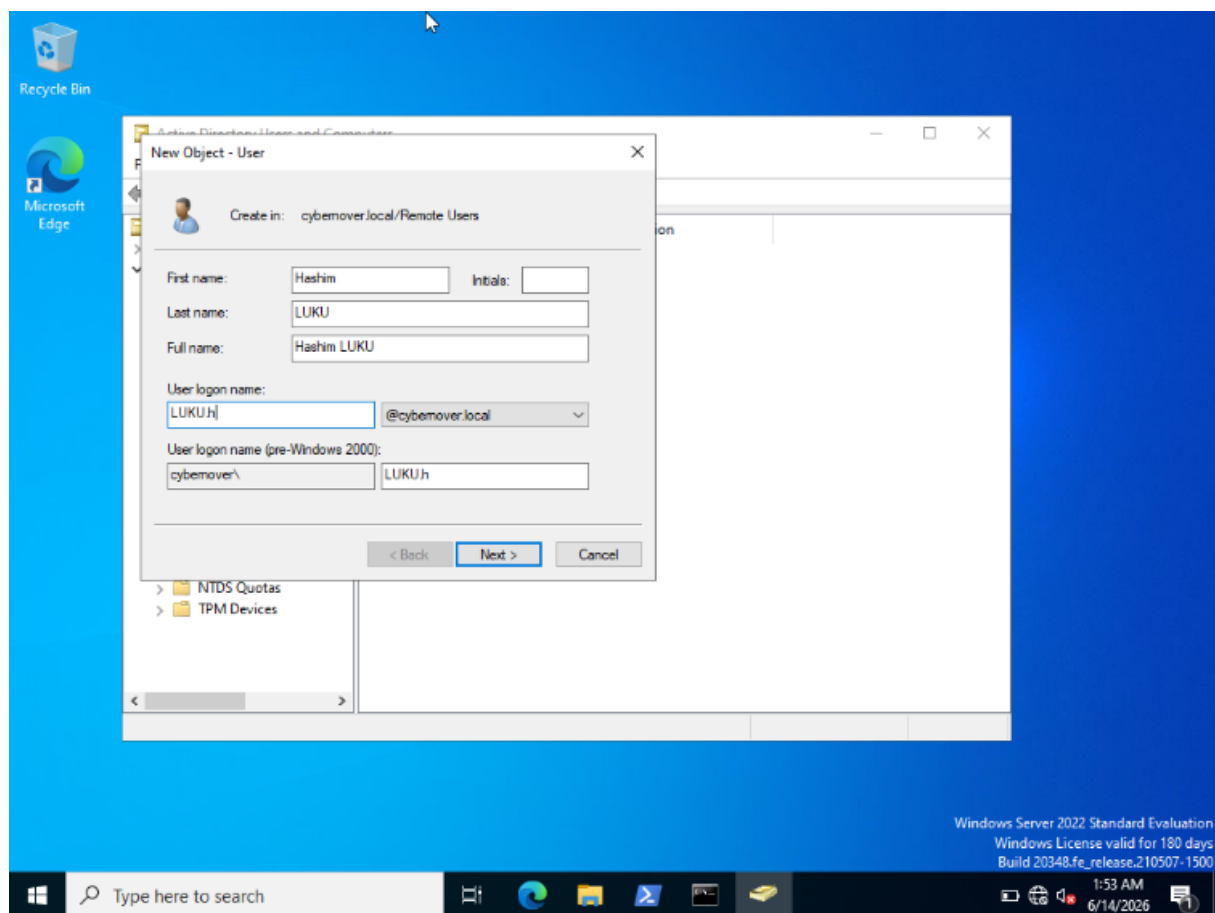


Figure 16: Operational Guide – User Management

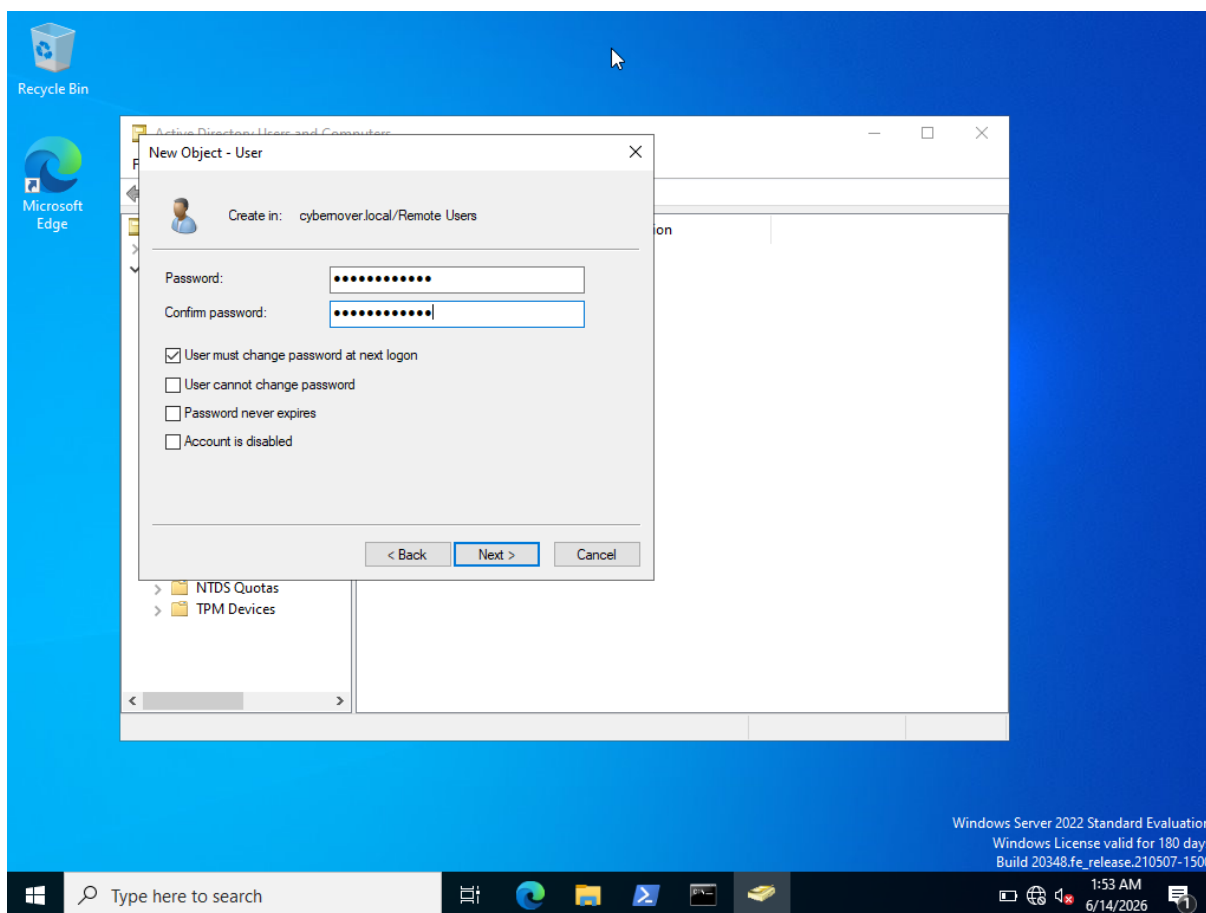


Figure 17: Operational Guide – Resource Access

12. Challenges Encountered and Solutions

Challenge	Solution
VirtualBox Guru Meditation error	Disabled Hyper-V on host, adjusted VM RAM/CPU settings
Network connectivity between VMs	Changed from NAT to Bridged Adapter for all VMs
Command Prompt restricted by GPO	Used PowerShell or Run dialog as alternative
Permission inheritance issues	Used Advanced Security settings to disable inheritance
Access Denied on file shares	Verified correct domain user login, not local account
Finance_Share creation blocked	Created DesktopShare as alternative; documented workaround
Name resolution (FS01 not found)	Used IP address \\172.16.1.20\ShareName

13. Conclusion

The Active Directory environment for CyberNova Technologies has been successfully deployed with all project requirements met. The following deliverables were completed:

- Domain Controller with cybernova.local domain
- 4 Departmental OUs (IT, HR, Finance, Remote Users)
- 4 Security Groups
- 9 User accounts across all departments
- 3 Group Policy Objects (Password/Lockout, User Restrictions)
- 3 Network Shares (IT_Share, HR_Share, DesktopShare for Finance)
- Proper NTFS and Share permissions (Principle of Least Privilege)
- Security hardening measures (Firewall, GPO restrictions, screen lock)

The solution meets all project requirements including centralized authentication, department-based access control, secure file sharing, and IT policy enforcement. The environment is fully functional and ready for production use by CyberNova Technologies.